

WTW 158 Worksheet 1 on inequalities and absolute value

A TEXTBOOK PROBLEMS

p A9: 17, 19, 27, 29, 35

p A9: 7, 12, 45, 49, 51, 55

B. MORE PROBLEMS - Solve: (Answer in interval notation)

1. $-3(r-4) \geq 0$

2. $m-2 < -8$ or $-\frac{m}{8} < -1$

3. $x+8 \geq 9$ and $2-x > -1$

4. $\frac{3-x}{x+1} \leq 0$

5. $\frac{|4x|}{x-3} < 0$

6. $\frac{1-x}{(x+2)(2-x)} \geq 0$

7. $\frac{x-1}{x} < 4$

8. $x|x| < 0$

9. p 39 Studyguide Technique mastering A - Absolute value

Do the following: 2 a, b, c, i, k, m

10. Write out in the form $f(x) = \begin{cases} \dots & \text{if } / \text{ as } x \in \dots \\ \dots & \text{if } / \text{ as } x \in \dots \end{cases}$ and sketch the function

(i) $f(x) = |4 - x^2|$

(ii) $f(x) = x|x|$

C. SELECTED TEST QUESTIONS

Question 1

Solve for x : $1 - \frac{1}{1-x} > 0$

Question 2

The solution set of $|x-1| > -1$ is

[2a] ϕ (empty set)	[2b] $\mathbb{R}$	[2c] $(0, \infty)$	[2d] $(-\infty, 2)$
[2e] $(-\infty, 0) \cup (2, \infty)$	[2f] $(0, 2)$	[2g] None of these	

Question 3

i Write down the definition of the absolute value of a real number x .

ii Write $\sqrt{|x-1|+4}$ in the form $\begin{cases} \dots & \text{if } x \geq \dots \\ \dots & \text{if } x < \dots \end{cases}$

iii Sketch the graph of the function $f(x) = \sqrt{|x-1|+4}$.

D. YOUTUBE VIDEOS - for extra exposure

Search:

1. Solving rational inequalities Mario's Math Tutoring

2. Solving Absolute Value Inequalities Mario's Math Tutoring

SELECTED ANSWERS

A. 17. $x \in (3, \infty)$, 19. $x \in (2, 6)$, 27. $x \in [-1, \frac{1}{2}]$, 35. $x \in (-1, 0) \cup (1, \infty)$

7. $2-x$, 45. $x \in \{2, -\frac{4}{3}\}$, 49. $x \in (3, 5)$, 51. $x \in \mathbb{R} \setminus (-7, -3)$

B. 1. $r \in (-\infty, 4]$, 2. $m \in \mathbb{R} \setminus [-6, 8]$, 3. $x \in [1, 3)$, 4. $x \in \mathbb{R} \setminus [-1, 3)$, 5. $x \in (-\infty, 3) \setminus \{0\}$,

6. $x \in (-2, 1] \cup (2, \infty)$, 7. $x \in \mathbb{R} \setminus [-\frac{1}{3}, 0]$, 8. $x \in (-\infty, 0)$, 9. 2a. $x = 2$, 2b. $x \neq 2$,

2c. $x \in \phi$, 2i. $x \in \{-1, 2\}$, 2k. $x \in \phi$, 2m. $x \in \{-3, 3\}$

A. TEXTBOOK PROBLEMS

PA9:

17. $2x+1 < 5x-8$

$\Rightarrow -3x < -9$

$\Rightarrow x > 3$

$\Rightarrow x \in (3, \infty)$

19. $-1 < 2x-5 < 7$

$\Rightarrow 4 < 2x < 12$

$\Rightarrow 2 < x < 6$

$\Rightarrow x \in (2, 6)$

27. $2x^2 + x \leq 1$

$\Rightarrow 2x^2 + x - 1 \leq 0$

$\Rightarrow (2x-1)(x+1) \leq 0$

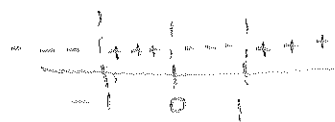
$\Rightarrow x \in [-1, \frac{1}{2}]$



35. $x^3 > x$

$\Rightarrow x^3 - x > 0$

$\Rightarrow x(x-1)(x+1) > 0$



$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

PA9:

7. $|x-2| = 2-x$ if $x \leq 2$

45. $|x+3| = |2x+1|$

$\Rightarrow x+3 = 2x+1$ or $x+3 = -(2x+1)$

$\Rightarrow x = 2$ or $x = -4/3$

$\Rightarrow x \in \{2, -4/3\}$

49. $|x-4| < 1$

$\Rightarrow -1 < x-4 < 1$

$\Rightarrow 3 < x < 5$

$\Rightarrow x \in (3, 5)$

51. $|x+5| \geq 2$

$\Rightarrow x+5 \geq 2$ or $x+5 \leq -2$

$\Rightarrow x \geq -3$ or $x \leq -7$

$\Rightarrow x \in \mathbb{R} \setminus (-7, -3)$

B. MORE PROBLEMS

1. $x-4 \leq 0$

$\Rightarrow x \leq 4$

$\Rightarrow x \in (-\infty, 4]$

2. $m < -6$ or $m > 8$

$\Rightarrow m \in \mathbb{R} \setminus [-6, 8]$

3. $x \geq 1$ and $x < 3$

$\Rightarrow x \in [1, 3)$

4. $\frac{3-x}{x+1} \leq 0$



$\Rightarrow x \in (-\infty, -1) \cup [3, \infty)$

$\Rightarrow x \in \mathbb{R} \setminus [-1, 3)$

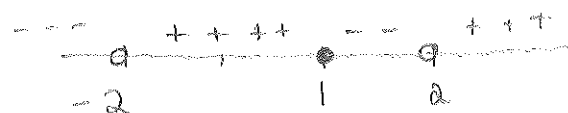
5. $\frac{|4x|}{x-3} < 0$

$\Rightarrow x-3 < 0$ and $x \neq 0$

$\Rightarrow x < 3$, $x \neq 0$

$\Rightarrow x \in (-\infty, 3) \setminus \{0\}$

6. $\frac{1-x}{(x+2)(2-x)} \geq 0$



$\Rightarrow x \in (-\infty, -2) \cup [1, 2) \cup (2, \infty)$

7. $\frac{x-1}{x} < 4$

$\Rightarrow \frac{x-1}{x} - 4 < 0$

$\Rightarrow \frac{x-1-4x}{x} < 0$

$\Rightarrow \frac{-3x-1}{x} < 0$

7. (continued)

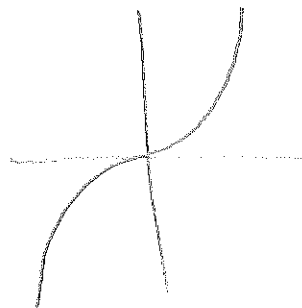
$$\frac{3x+1}{x} > 0 \Rightarrow (3x+1)x > 0$$



$$\Rightarrow x \in (-\infty, -\frac{1}{3}) \cup (0, \infty)$$

$$\Rightarrow x \in \mathbb{R} \setminus [-\frac{1}{3}, 0]$$

$$(ii) f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$



C. TEST QUESTIONS

8. $x|x| < 0$

$$\Rightarrow x < 0$$

$$\Rightarrow x \in (-\infty, 0)$$

9. p 39

29. a. $x = 2$ b. $x \neq 2$

c. $x \in \emptyset$ d. $x \in \mathbb{R}$

e. $1-2x=3$ or $1-2x=-3$

$$\Rightarrow x = -1 \text{ or } x = 2$$

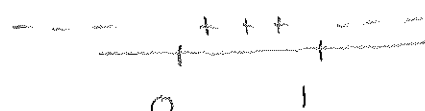
k. $x \in \emptyset$ m. $|x|=3$

$$\Rightarrow x = 3 \text{ or } x = -3$$

1. $\frac{1-x-1}{1-x} > 0$

$$\Rightarrow \frac{-x}{1-x} > 0$$

$$\Rightarrow \frac{x}{1-x} < 0 \Rightarrow (x(1-x) < 0)$$



$$\Rightarrow x \in (-\infty, 0) \cup (1, \infty)$$

$$\Rightarrow x \in \mathbb{R} \setminus [0, 1]$$

2. b.

$$3(i) |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

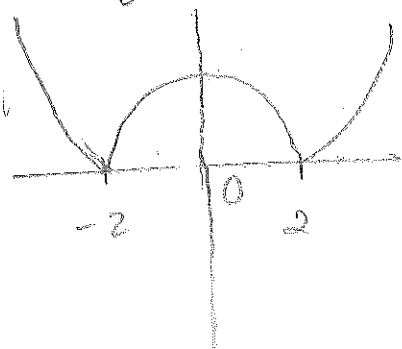
10. (i) $f(x) = \begin{cases} 4-x^2 & \text{if } 4-x^2 \geq 0 \\ -(4-x^2) & \text{if } 4-x^2 < 0 \end{cases}$

$$= \begin{cases} 4-x^2 & \text{if } x^2 \leq 4 \\ x^2-4 & \text{if } x^2 > 4 \end{cases}$$

$$= \begin{cases} 4-x^2 & \text{if } x \in [-2, 2] \\ x^2-4 & \text{if } x \in \mathbb{R} \setminus (-2, 2) \end{cases}$$

$$= \begin{cases} 4-x^2 & \text{if } x \in [-2, 2] \\ x^2-4 & \text{if } x \in \mathbb{R} \setminus (-2, 2) \end{cases}$$

$$= \begin{cases} 4-x^2 & \text{if } x \in [-2, 2] \\ x^2-4 & \text{if } x \in \mathbb{R} \setminus (-2, 2) \end{cases}$$



(ii) $\sqrt{|x-1|+4}$

$$= \sqrt{x+3} \text{ if } x-1 \geq 0$$

$$= \sqrt{-x+5} \text{ if } x-1 < 0$$

$$= \begin{cases} \sqrt{x+3} & \text{if } x \geq 1 \\ \sqrt{-x+5} & \text{if } x < 1 \end{cases}$$

$$= \begin{cases} \sqrt{x+3} & \text{if } x \geq 1 \\ \sqrt{-x+5} & \text{if } x < 1 \end{cases}$$

